

Grade 8
Unit 2
Lesson 5 Practice

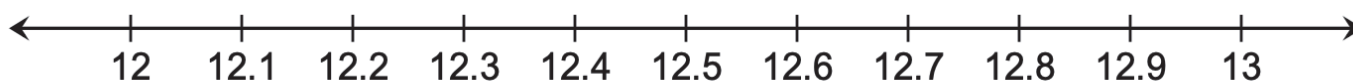
Name _____
Date _____
Period _____

For each of the values below in questions 1-5, identify whether it is rational or irrational.

1. $\pi + 5.5$ is a ☐ rational.
☐ irrational. number because it ☐ can
☐ cannot be expressed as a ratio of two integers.
2. $-\sqrt{135}$ is a ☐ rational.
☐ irrational. number because it ☐ can
☐ cannot be expressed as a ratio of two integers.
3. $\sqrt[3]{-512}$ is a ☐ rational.
☐ irrational. number because it ☐ can
☐ cannot be expressed as a ratio of two integers.
4. $-12.12\bar{3}$ is a ☐ rational.
☐ irrational. number because it ☐ can
☐ cannot be expressed as a ratio of two integers.
5. $0.5435432 \dots$ is a ☐ rational.
☐ irrational. number because it ☐ can
☐ cannot be expressed as a ratio of two integers.

Consider the expression $\sqrt{150}$.

6. Between which two whole numbers does $\sqrt{150}$ lie?
7. To the nearest tenth, approximate the value of $\sqrt{150}$. Then plot the approximate location on the number line below.



8. Use the approximation of $\sqrt{150}$ to approximate the value of each expression in the table.

Expression	Approximate Value
$12 - \sqrt{150}$	
$2\sqrt{150}$	
$\frac{\sqrt{150}}{6}$	

9. Approximate the value of each expression.

	Expression	Approximate Value
A	$\sqrt[3]{-198}$	
B	$\frac{1}{2}\sqrt[3]{-198}$	
C	$-\frac{3}{4}\pi$	
D	$\pi - 7$	

10. Plot and label the approximate location of each expression from question 9 using letters A, B, C, and D on the number line.

